Table 1: Nonlinear ranking results in the verified 20-entry unique-score universe.

One-sample - Additive sinusoidal

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.701	19	0.348	19	100
uPU logistic	0.702	17.5	0.348	17.5	100
nnPU logistic	0.702	17.5	0.348	17.5	100
Naive/Corrected AdaBoost	0.752	7	0.432	6	100
Naive/Corrected GBM	0.754	5	0.426	10	100
Naive/Corrected XGBoost	0.758	2	0.442	2	100
Naive/Corrected LightGBM	0.752	9	0.428	9	100
Naive/Corrected CatBoost	0.751	10	0.431	8	100
RN AdaBoost	0.718	13	0.372	14	100
RN GBM	0.708	16	0.359	15	100
RN XGBoost	0.715	14	0.375	13	100
RN LightGBM	0.701	20	0.344	20	100
RN CatBoost	0.711	15	0.358	16	100
CS AdaBoost	0.752	8	0.432	5	100
CS GBM	0.756	3	0.439	3	100
CS XGBoost	0.759	1	0.444	1	100
CS LightGBM	0.752	6	0.432	7	100
CS CatBoost	0.754	4	0.438	4	100
AdaPU	0.723	12	0.381	12	100
PU gradient boosting	0.739	11	0.412	11	100

One-sample - Radial / concentric

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.501	15	0.246	15	100
uPU logistic	0.501	13.5	0.247	13.5	100
nnPU logistic	0.501	13.5	0.247	13.5	100
Naive/Corrected AdaBoost	0.783	11	0.586	11	100
Naive/Corrected GBM	0.784	7	0.601	7	100
Naive/Corrected XGBoost	0.787	5	0.608	4	100
Naive/Corrected LightGBM	0.784	9	0.597	9	100
Naive/Corrected CatBoost	0.789	1	0.617	1	100
RN AdaBoost	0.485	16	0.205	16	100
RN GBM	0.465	20	0.192	20	99
RN XGBoost	0.473	19	0.198	19	100
RN LightGBM	0.481	17	0.203	17	100
RN CatBoost	0.474	18	0.200	18	100
CS AdaBoost	0.784	8	0.589	10	100
CS GBM	0.786	6	0.606	6	100
CS XGBoost	0.788	4	0.607	5	100
CS LightGBM	0.783	10	0.598	8	100
CS CatBoost	0.789	2	0.616	2	100
AdaPU	0.789	3	0.612	3	100
PU gradient boosting	0.768	12	0.582	12	100

One-sample - Pure XOR interaction

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.500	17	0.220	16	100

One-sample - Pure XOR interaction

Ranking entry	AUROC	AUC rank	AP	AP rank	n
uPU logistic	0.500	18.5	0.220	17.5	100
nnPU logistic	0.500	18.5	0.220	17.5	100
Naive/Corrected AdaBoost	0.540	12	0.238	11	100
Naive/Corrected GBM	0.753	8	0.444	8	100
Naive/Corrected XGBoost	0.787	4	0.499	3	100
Naive/Corrected LightGBM	0.772	6	0.477	6	100
Naive/Corrected CatBoost	0.829	2	0.580	1	100
RN AdaBoost	0.524	16	0.221	15	100
RN GBM	0.525	15	0.215	19	100
RN XGBoost	0.529	14	0.221	14	100
RN LightGBM	0.537	13	0.226	13	100
RN CatBoost	0.543	11	0.235	12	100
CS AdaBoost	0.544	10	0.242	10	100
CS GBM	0.763	7	0.466	7	100
CS XGBoost	0.789	3	0.496	4	100
CS LightGBM	0.776	5	0.481	5	100
CS CatBoost	0.829	1	0.578	2	100
AdaPU	0.500	20	0.202	20	100
PU gradient boosting	0.580	9	0.290	9	100

One-sample - Threshold interaction

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.501	14	0.201	12	100
uPU logistic	0.501	12.5	0.201	13.5	100
nnPU logistic	0.501	12.5	0.201	13.5	100
Naive/Corrected AdaBoost	0.499	19	0.200	19	100
Naive/Corrected GBM	0.537	8	0.221	8	100
Naive/Corrected XGBoost	0.548	6	0.233	6	100
Naive/Corrected LightGBM	0.583	4	0.260	4	100
Naive/Corrected CatBoost	0.669	2	0.334	2	100
RN AdaBoost	0.500	16	0.200	18	100
RN GBM	0.502	11	0.201	15	100
RN XGBoost	0.501	15	0.201	16	100
RN LightGBM	0.503	10	0.201	11	100
RN CatBoost	0.503	9	0.202	9	100
CS AdaBoost	0.499	20	0.200	20	100
CS GBM	0.545	7	0.229	7	100
CS XGBoost	0.550	5	0.233	5	100
CS LightGBM	0.583	3	0.261	3	100
CS CatBoost	0.679	1	0.341	1	100
AdaPU	0.500	17	0.201	17	100
PU gradient boosting	0.500	18	0.202	10	100

One-sample - Sinusoidal + 497 noise predictors ($p = 500$)

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.609	20	0.272	20	100
uPU logistic	0.622	18.5	0.282	18.5	100
nnPU logistic	0.622	18.5	0.282	18.5	100

One-sample - Sinusoidal + 497 noise predictors ($p = 500$)

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Naive/Corrected AdaBoost	0.737	5	0.399	8	100
Naive/Corrected GBM	0.736	6	0.390	11	100
Naive/Corrected XGBoost	0.740	2	0.410	2	100
Naive/Corrected LightGBM	0.734	9	0.397	9	100
Naive/Corrected CatBoost	0.734	10	0.400	7	100
RN AdaBoost	0.730	11	0.390	10	100
RN GBM	0.725	14	0.380	14	100
RN XGBoost	0.725	13	0.381	13	100
RN LightGBM	0.718	17	0.370	17	100
RN CatBoost	0.726	12	0.387	12	100
CS AdaBoost	0.738	4	0.402	6	100
CS GBM	0.740	3	0.407	3	100
CS XGBoost	0.741	1	0.413	1	100
CS LightGBM	0.735	7	0.403	4	100
CS CatBoost	0.735	8	0.403	5	100
AdaPU	0.722	15	0.379	15	100
PU gradient boosting	0.721	16	0.374	16	100

One-sample - Nonlinear heterogeneous covariance

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.499	20	0.210	15	100
uPU logistic	0.499	18.5	0.210	17.5	100
nnPU logistic	0.499	18.5	0.210	17.5	100
Naive/Corrected AdaBoost	0.564	9	0.250	10	100
Naive/Corrected GBM	0.736	8	0.415	8	100
Naive/Corrected XGBoost	0.779	4	0.501	4	100
Naive/Corrected LightGBM	0.763	6	0.479	6	100
Naive/Corrected CatBoost	0.819	2	0.574	1	100
RN AdaBoost	0.513	16	0.209	19	100
RN GBM	0.519	15	0.209	20	100
RN XGBoost	0.521	14	0.210	14	100
RN LightGBM	0.526	13	0.214	13	100
RN CatBoost	0.527	12	0.219	12	100
CS AdaBoost	0.561	10	0.248	11	100
CS GBM	0.749	7	0.452	7	100
CS XGBoost	0.782	3	0.505	3	100
CS LightGBM	0.764	5	0.483	5	100
CS CatBoost	0.820	1	0.573	2	100
AdaPU	0.510	17	0.210	16	100
PU gradient boosting	0.560	11	0.275	9	100

Two-sample - Additive sinusoidal

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.703	18	0.347	20	100
uPU logistic	0.703	19.5	0.348	18.5	100
nnPU logistic	0.703	19.5	0.348	18.5	100
Naive/Corrected AdaBoost	0.751	7	0.428	6	100
Naive/Corrected GBM	0.753	5	0.422	9	100

Two-sample - Additive sinusoidal

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Naive/Corrected XGBoost	0.757	1	0.438	2	100
Naive/Corrected LightGBM	0.750	9	0.422	10	100
Naive/Corrected CatBoost	0.749	10	0.426	8	100
RN AdaBoost	0.718	13	0.367	14	100
RN GBM	0.710	16	0.358	15	100
RN XGBoost	0.717	14	0.374	13	100
RN LightGBM	0.705	17	0.348	17	100
RN CatBoost	0.712	15	0.357	16	100
CS AdaBoost	0.751	6	0.428	5	100
CS GBM	0.755	3	0.433	3	100
CS XGBoost	0.757	2	0.439	1	100
CS LightGBM	0.750	8	0.426	7	100
CS CatBoost	0.753	4	0.433	4	100
AdaPU	0.726	12	0.382	12	100
PU gradient boosting	0.743	11	0.414	11	100

Two-sample - Radial / concentric

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.499	13	0.241	15	100
uPU logistic	0.499	14.5	0.246	13.5	100
nnPU logistic	0.499	14.5	0.246	13.5	100
Naive/Corrected AdaBoost	0.782	10	0.584	11	100
Naive/Corrected GBM	0.785	7	0.598	7	100
Naive/Corrected XGBoost	0.786	5	0.604	4	100
Naive/Corrected LightGBM	0.782	9	0.587	9	100
Naive/Corrected CatBoost	0.789	1	0.616	1	100
RN AdaBoost	0.480	16	0.201	16	99
RN GBM	0.467	20	0.193	20	99
RN XGBoost	0.472	19	0.197	19	100
RN LightGBM	0.477	17	0.200	17	100
RN CatBoost	0.473	18	0.200	18	99
CS AdaBoost	0.784	8	0.587	10	100
CS GBM	0.785	6	0.601	6	100
CS XGBoost	0.786	4	0.602	5	100
CS LightGBM	0.782	11	0.589	8	100
CS CatBoost	0.789	2	0.614	2	100
AdaPU	0.788	3	0.609	3	100
PU gradient boosting	0.764	12	0.563	12	100

Two-sample - Pure XOR interaction

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.501	19	0.218	19	100
uPU logistic	0.501	17.5	0.220	14.5	100
nnPU logistic	0.501	17.5	0.220	14.5	100
Naive/Corrected AdaBoost	0.534	13	0.230	12	100
Naive/Corrected GBM	0.735	8	0.413	8	100
Naive/Corrected XGBoost	0.781	4	0.482	3	100
Naive/Corrected LightGBM	0.768	6	0.465	6	100

Two-sample - Pure XOR interaction

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Naive/Corrected CatBoost	0.829	1	0.576	1	100
RN AdaBoost	0.522	16	0.219	18	99
RN GBM	0.527	15	0.219	17	99
RN XGBoost	0.528	14	0.220	16	100
RN LightGBM	0.534	12	0.224	13	100
RN CatBoost	0.541	11	0.233	11	100
CS AdaBoost	0.542	10	0.238	10	100
CS GBM	0.744	7	0.432	7	100
CS XGBoost	0.784	3	0.481	4	100
CS LightGBM	0.769	5	0.467	5	100
CS CatBoost	0.829	2	0.574	2	100
AdaPU	0.499	20	0.201	20	100
PU gradient boosting	0.581	9	0.272	9	100

Two-sample - Threshold interaction

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.501	14	0.202	12	100
uPU logistic	0.501	15.5	0.202	13.5	100
nnPU logistic	0.501	15.5	0.202	13.5	100
Naive/Corrected AdaBoost	0.500	20	0.201	20	100
Naive/Corrected GBM	0.530	8	0.218	8	100
Naive/Corrected XGBoost	0.547	5	0.232	5	100
Naive/Corrected LightGBM	0.571	4	0.251	4	100
Naive/Corrected CatBoost	0.656	2	0.320	2	100
RN AdaBoost	0.500	18	0.201	18	100
RN GBM	0.502	12	0.202	16	100
RN XGBoost	0.501	13	0.202	15	100
RN LightGBM	0.503	10	0.202	11	100
RN CatBoost	0.505	9	0.204	9	100
CS AdaBoost	0.500	17	0.201	19	100
CS GBM	0.537	7	0.224	7	100
CS XGBoost	0.547	6	0.231	6	100
CS LightGBM	0.574	3	0.253	3	100
CS CatBoost	0.669	1	0.331	1	100
AdaPU	0.500	19	0.201	17	100
PU gradient boosting	0.503	11	0.202	10	100

Two-sample - Sinusoidal + 497 noise predictors ($p = 500$)

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.604	19	0.267	19	100
uPU logistic	0.591	20	0.258	20	100
nnPU logistic	0.609	18	0.271	18	100
Naive/Corrected AdaBoost	0.735	3	0.395	4	100
Naive/Corrected GBM	0.731	6	0.377	12	100
Naive/Corrected XGBoost	0.735	4	0.398	2	100
Naive/Corrected LightGBM	0.728	11	0.384	10	100
Naive/Corrected CatBoost	0.729	10	0.391	8	100
RN AdaBoost	0.729	9	0.387	9	100

Two-sample - Sinusoidal + 497 noise predictors ($p = 500$)

Ranking entry	AUROC	AUC rank	AP	AP rank	n
RN GBM	0.721	16	0.372	16	100
RN XGBoost	0.724	13	0.376	14	100
RN LightGBM	0.715	17	0.364	17	100
RN CatBoost	0.723	14	0.380	11	100
CS AdaBoost	0.736	2	0.397	3	100
CS GBM	0.735	5	0.393	5	100
CS XGBoost	0.737	1	0.403	1	100
CS LightGBM	0.730	7	0.391	7	100
CS CatBoost	0.730	8	0.392	6	100
AdaPU	0.721	15	0.377	13	100
PU gradient boosting	0.724	12	0.373	15	100

Two-sample - Nonlinear heterogeneous covariance

Ranking entry	AUROC	AUC rank	AP	AP rank	n
Logistic probability correction	0.501	18	0.209	19	100
uPU logistic	0.501	19.5	0.209	16.5	100
nnPU logistic	0.501	19.5	0.209	16.5	100
Naive/Corrected AdaBoost	0.563	11	0.250	10	100
Naive/Corrected GBM	0.727	8	0.396	8	100
Naive/Corrected XGBoost	0.768	4	0.481	4	100
Naive/Corrected LightGBM	0.750	6	0.456	6	100
Naive/Corrected CatBoost	0.814	2	0.562	2	100
RN AdaBoost	0.511	17	0.207	20	100
RN GBM	0.519	14	0.209	18	100
RN XGBoost	0.518	15	0.210	14	100
RN LightGBM	0.523	13	0.214	13	100
RN CatBoost	0.526	12	0.220	12	100
CS AdaBoost	0.564	10	0.250	11	100
CS GBM	0.736	7	0.430	7	100
CS XGBoost	0.770	3	0.484	3	100
CS LightGBM	0.753	5	0.461	5	100
CS CatBoost	0.816	1	0.563	1	100
AdaPU	0.511	16	0.210	15	100
PU gradient boosting	0.595	9	0.291	9	100

Note. Twenty unique ranking entries per nonlinear DGP/design. Verified naive/corrected same-engine aliases retain one score without averaging. Ranks use the saved unique-score analysis mean-score rule, with average ties after 12-decimal rounding. n=99 entries are conditional on successful refits. Original 25-configuration rank files remain unchanged. The original benchmark two-sample AdaBoost exception remains separate in its 21-entry ranking universe; it does not apply to these nonlinear aliases.